Sahaj Bhadja

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Summary

Backend software engineer with 2+ years owning **education platforms, event pipelines, and system integrations**; shipped reliable systems handling 2M+ records and 10K+ daily requests through data reconciliation, failure isolation, observability, testing, and production automation.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Owned **backend and data infrastructure for an education platform**, designing normalized schemas, row-level isolation, and indexed access patterns for 500+ users while reducing dashboard latency from 1.2 seconds to 350 milliseconds.
- Built resilient ingestion and reconciliation workflows for 5,000+ monthly jobs, handling malformed data, incremental loads, schema-aware validation, explicit states, and failure isolation to maintain 99%+ production reliability through CI/CD.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Shipped a **production education application** serving 150+ instructors and 10,000+ daily requests, building Flask services for secure uploads, processing, history, and reporting while reducing user turnaround time 60%.
- Debugged unfamiliar integrations across multiple AI and OCR providers, reproducing file-specific failures and implementing validation, persistent results, fallback handling, and automated tests to keep concurrent processing stable under peak load.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Owned **cross-platform installation, build, and validation workflows** across Windows, Linux, and macOS, automating nightly execution, failure reports, and 20+ tests to make production releases repeatable and independently supportable.
- Diagnosed and optimized API and ETL components using SQL, caching, AWS Glue, and S3, reducing response times 30%, pipeline processing 68%, storage costs 15%, and recurring manual effort 35%.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built **real-time event and synchronization services** across eight Node.js components and 2M+ records, supporting authentication, resource assignment, permissions, notifications, and dashboard updates through REST APIs, WebSockets, and Azure SQL.
- Used Sentry, Azure Monitor, Kibana, and Google Analytics to distinguish application, data, and usage failures, reducing synchronization conflicts 33% and shipping tested Docker releases through Jenkins 50% faster.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Developed **low-latency event-driven services** for bookings, pricing, payments, and status changes across 50,000+ records, using Firebase transactions, real-time synchronization, and validated webhooks to maintain sub-200-millisecond API responses.
- Made transaction workflows idempotent and reconciliation-driven across refunds, disputes, and payout updates, reducing manual payment handling 70% and eliminating repeated writes while improving booking confirmation speed 40%.

Skills

Languages: Python, Java, JavaScript, TypeScript, SQL, C++, Bash
Backend: FastAPI, Flask, Spring Boot, Node.js, REST APIs, WebSockets, event pipelines, microservices
Data: PostgreSQL, SQL Server, Azure SQL, MongoDB, Firebase, schema design, reconciliation, idempotency
Reliability: Distributed systems, concurrency, failure isolation, observability, Sentry, Azure Monitor, Kibana, testing
Delivery & AI: AWS, Azure, Docker, Kubernetes, Git, Jenkins, CI/CD, LLM orchestration, prompt chains, AI output validation

Projects

FinNexus Live | *Event Systems, MongoDB, Kubernetes*
- Built a **real-time financial event platform** supporting 50,000+ users and 1M+ transactions with secure bank integrations, token-based access, synchronization, structured filtering, MongoDB persistence, and Kubernetes-based production automation.

Chat2SQL Code | *Java, TypeScript, SQL, Docker, Maven*
- Enhanced an **AI-assisted data integration application** supporting 10+ SQL/NoSQL systems with schema exploration, structure synchronization, query generation, reusable interfaces, Docker, Maven, and cross-platform troubleshooting workflows across development environments.

IoT Authentication Analysis | *IoT, Device Security, Access Control*
- Designed and evaluated **secure IoT device authentication** across three user scenarios and four permission tiers, replacing centralized verification with tamper-resistant identity workflows while reducing repeated key-verification steps 50%.